An Integrated Approach to Addressing Multiple SDG Challenge

**Securing Safe Water and Improving Sanitation,
Strengthening Health and Livelihoods through Agroforestry,
Reducing Child Labor, and Promoting CO_2 Sequestration**

12 Sep 2026 Kenichi Miyauchi

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1. Introduction

1-1. About This Proposal

This proposal takes an integrated approach to complex issues such as global water problems, poverty, child labor, and CO_2 emissions.

- With less than five years remaining until the 2030 SDG deadline, many challenges still remain.
- Many of these challenges are linked to the lack of safe, sanitary water.
- Increasing CO_2 sequestration can contribute to climate change mitigation and help reduce climate-related risks.

The factors behind the issues addressed by the SDGs are intricately interconnected.

Water & Sanitation

- Global water issues
- Contaminated water
- Water collection (lost education)

Society & Economy

- Poverty
- Child labor issues
- Unable to attend school

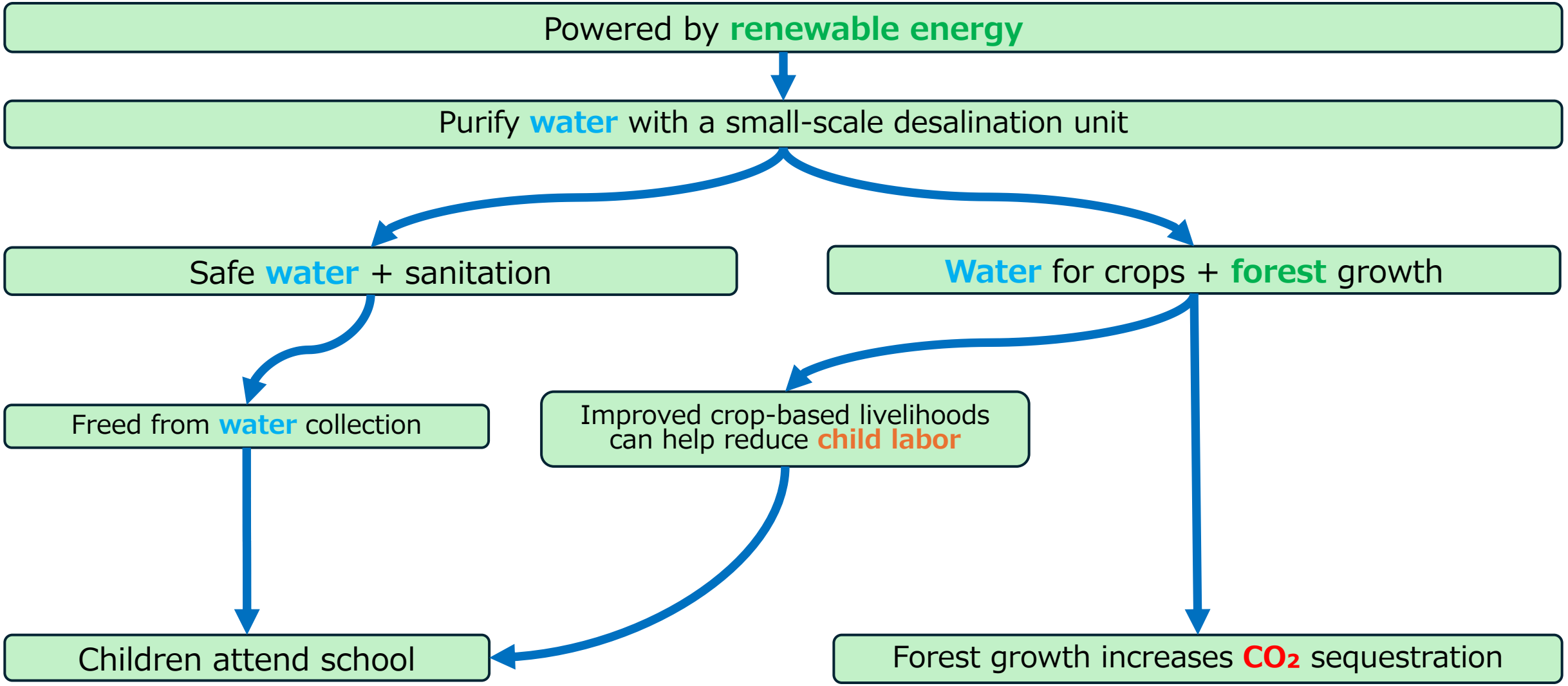
Agriculture & Environment

- Food shortages
- Climate vulnerability
- CO_2 reduction

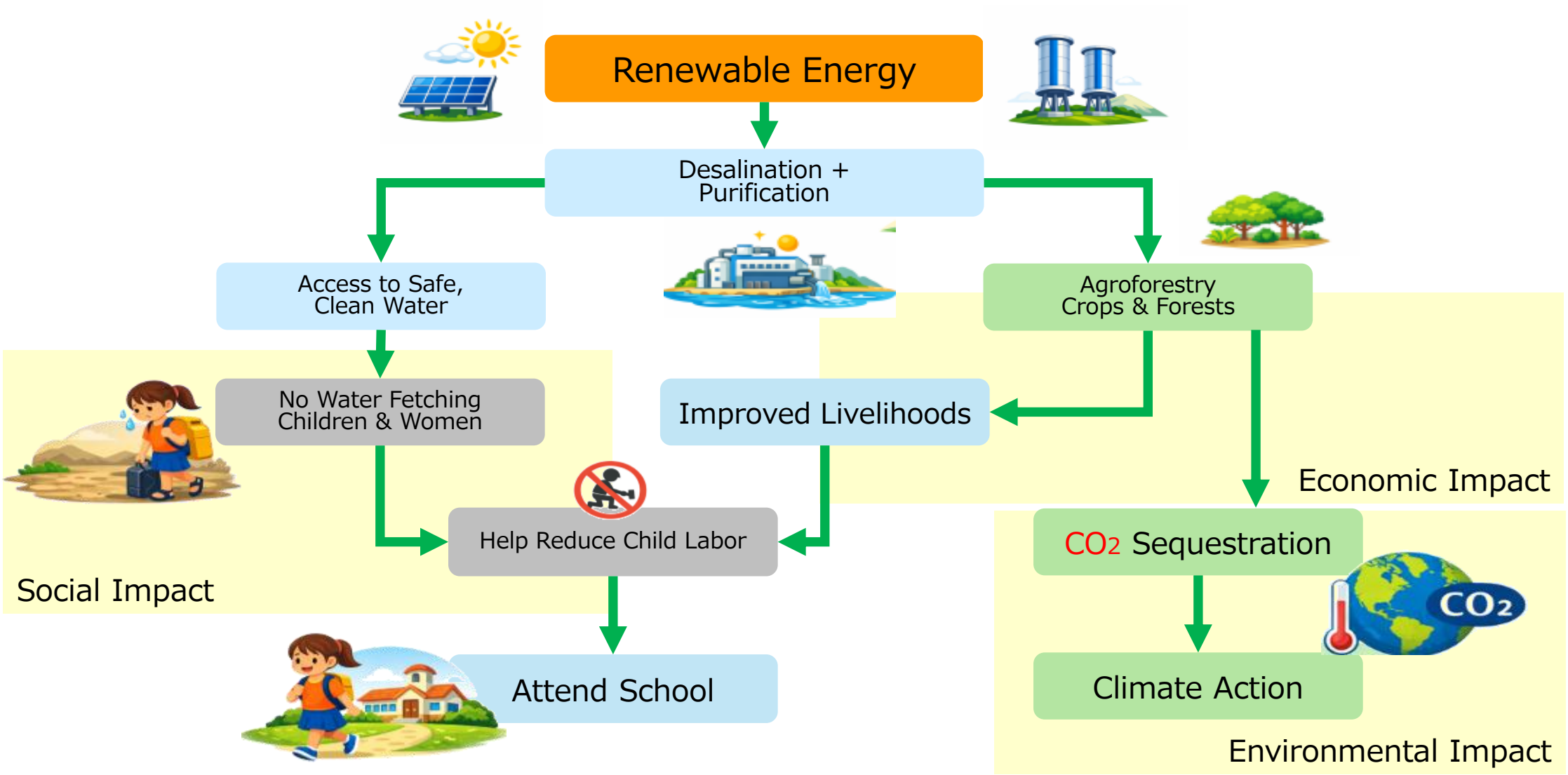
1-2. Key Technologies

Category	Technology	Impact Enabled
Infrastructure	Small-scale desalination unit Waqua MYZ-E250	Access to safe, sanitary water Safe toilets with lighting Freedom from water collection Water transport to inland areas
Health	Basic healthcare / Primary healthcare	Improved access to essential health services
Energy	Renewable energy Silicon photovoltaics Perovskite photovoltaics Biodiesel	Power source for desalination
Agriculture & Environment	Agroforestry	Livelihood improvement and local development through crops Helps reduce child labor and support school attendance CO_2 sequestration and greening

1-2. SDGs: Overall View of a Simultaneous Approach to Multiple Issues



1-3. SDGs: Illustration of a Simultaneous Approach to Multiple Issues



1-3. SDGs: Illustration of a Simultaneous Approach to Multiple Issues

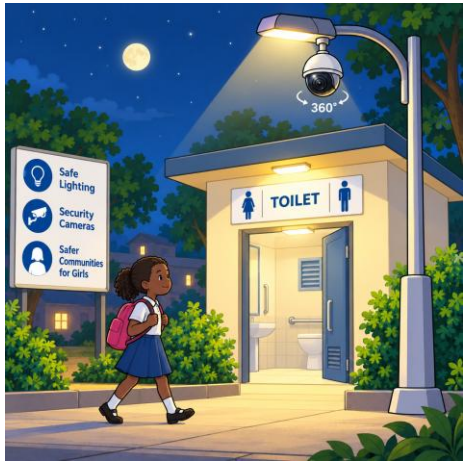
(1) Access to Basic Healthcare

Improving access to essential healthcare services for children and families.



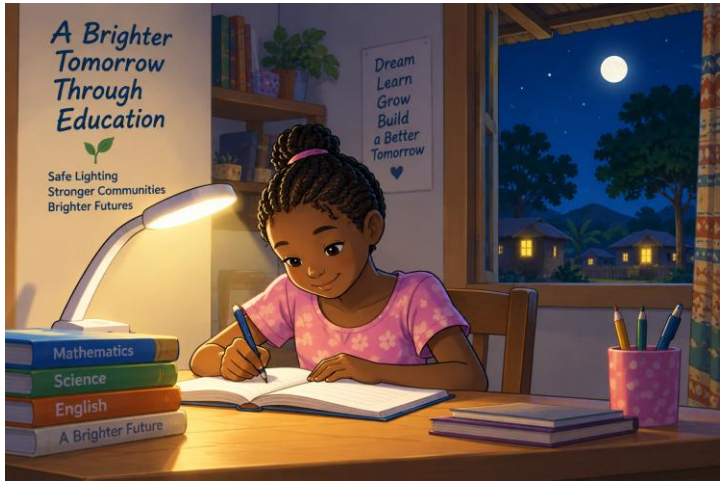
(2) Safe Toilets and Nighttime Safety

Enabling girls and other children to safely use toilets at night.



(3) Lighting for Nighttime Study

Enabling children to study safely at night with LED lighting.



1-4. Environmental Regeneration Effects of Agroforestry

- Climate Change Mitigation (Increasing CO_2 Sequestration)
  - Increase CO_2 sequestration through forest growth and agroforestry.
  - Contribute to climate change mitigation and help reduce climate-related risks.

- Regeneration of Regional Ecosystems
  - May contribute to changes in local water-cycle dynamics through increased forest cover.
  - Improve dryland climate adaptation by greening the Sahel region.

2. About Water

2-1. Water Requirements (Drinking Water + Domestic Water)

- UNICEF Reference level

  15 L/person/day　　of which drinking water is 3–5 L

  https://www.unicef.or.jp/library/pres_bn2007/pres_07_33.html

- Assumption in This Model

  30 L/person/day　Amount assuming improved living standards.

- Water required for a settlement of 1,000 people

  30 L × 1,000 people = 30,000 L/day

2-2. Water Requirements (Crops + Forest Growth)

Water required for crops and forests

Estimated water requirement per hectare for dryland agroforestry: **8,000–16,000 L/day/ha**
Based on CIFOR (Center for International Forestry Research) dryland studies
and technical documents such as “Agroforestry and Water Management in Drylands.”
Reference: CIFOR Publications (https://www.cifor.org/knowledge/publications/)
Considering evaporation on extremely hot days, salt accumulation prevention,
and climate-change risk (buffer), this model adopts **14,250 L/day/ha** as a design value.
When cultivated land is 20–40 ha, a large amount of water is required.
To avoid higher installation costs for small-scale desalination units,
adopt the water-saving technologies on the next page.

2-3. Water-Saving Technologies

Optimizing Water Retention for Crops and Forests

An approach to prevent excessive soil absorption and evaporation by “narrowing the outlet,” “covering the surface,” and “increasing water retention.”

- **Drip Irrigation**
- **Mulching (ground cover)**
- Use of water-retaining materials and soil conditioners
- Placement of windbreak forests (windbreaks)
- Subsurface irrigation (watering underground)

2-4. Water Requirements (1,000-person Settlement + 40 ha)

- Drinking & domestic water (1,000 people): 30,000 L/day
- Agroforestry water (40 ha): 570,000 L/day
  *Optimized by applying water-saving technologies (drip irrigation + mulching)

- Total daily water requirement 600,000 L/day
- Supply Plan
  - Water storage tank capacity: 900,000 L
    Always secure 1.5 days of supply; stable even during weather changes and maintenance
  - Transport system: Supply each inland area from desalination hubs using biodiesel water trucks

[Supplement: Required water by farmland area]

1 ha: 14,250 L / 20 ha: 285,000 L / 30 ha: 427,500 L / 40 ha: 570,000 L

3. About Seawater Desalination Systems

3-1. Small-Scale Seawater Desalination Unit



Purifies seawater and contaminated (raw) water.

- Desalination of seawater
- Purification of river and contaminated water through filtration systems
- Purification using RO (reverse osmosis) membranes
- Removes pathogens, viruses, and other contaminants
- Can remove microplastics
- Provides safe, treated water
- Stable supply through distributed deployment due to compact size
- Track record in overseas water and sanitation projects

3-1. Small-Scale Seawater Desalination Unit

[Secondary Effects]

This unit uses RO reverse-osmosis membranes and removes fine particles in seawater with high efficiency.
The current process has demonstrated the potential to remove more than 99% of microplastics in the 1–5 μm size range.

[Future Development Potential]

By adding a dedicated module applying RO reverse-osmosis membrane technology, microplastic removal from seawater could be further enhanced.
This suggests potential for further investigation as an approach to addressing marine pollution.

[Application to Inland Water Sources]

Although this unit is designed primarily as a small-scale seawater desalination system, it can also be used to purify freshwater sources such as river water, depending on water quality and pretreatment requirements.

4. About Renewable Energy

4-1. About Renewable Energy

- Silicon Photovoltaics
  - Conventional solar power technology
  - Based on the widely used 400 W panel.
  - Use salt-damage-resistant, corrosion-resistant panels

- Perovskite Photovoltaics
  - A solar power technology expected to be deployed in the future
  - Can generate power when applied to cylinders or glass surfaces
  - Could become more cost-competitive than silicon photovoltaics in the future

5. Small-Scale Seawater Desalination and Required Number of Solar Panels

5-1. Required Units (MYZ-E250) and Solar Panels

	Seawater	Surface Water
Daily Water Demand	**600,000 L ***	**600,000 L ***
Solar Power Generation Hours	7:00-17:00(10h)	7:00-17:00(10h)
Water Production (10h/day)	2300 L/10hr	4600 L/10hr
Number of Units Required	Approx. 260 Units	Approx. 130 Units
Operating Voltage	Single-phase 200V	Single-phase 200V
Power Consumption	3,810 W(Unit+Pump) 3,360 W(Unit Only)	1,990 W(Unit+Pump) 1,540 W(Unit Only)
Required Number of Solar Panels (400 W) (Upper: Rated, Lower: 70% Efficiency)	Approx. 2,480 panels Approx. 3,540 panels	Approx. 650 panels Approx. 930 panels

* Slide 13: 2-4: Water requirements for 1,000 people and 40 ha.

"260 units" is required due to the high pre-treatment load for seawater.
Figures are based on the current model (MYZ E-250).
Appendix cost estimates adopt the 70% efficiency figures (Seawater: 3,540, Surface Water: 930).

5-2. Selection of Water Intake Sites and Water Supply

- Selection Criteria for Water Intake Sites
  - Physical requirements: Calm sea (seawater desalination) or rivers, lakes, aquifers
  - Sustainability: Stable intake volume throughout the year (minimum: 600 m^3/day)
  - Social requirements: Sufficient consensus-building with local residents

- Delivery & Logistics (areas where securing water sources is difficult)
  - Securing delivery methods:
    - Current: Biodiesel water trucks
    - Future: EV water trucks equipped with sodium-ion batteries

6. Agroforestry: Crops and Trees

6-1. Agroforestry Structure and Value

Perspective	Role of Agroforestry	Economic & Environmental Value Created
Climate Resilience	Trees shade intense sunlight and moderate microclimates	Helps reduce the risk of crop loss during droughts (risk diversification)
Soil Environment	Leaf litter and roots add organic matter and nurture soil	Reduces chemical fertilizer dependence and improves long-term soil fertility
Revenue Structure	Multi-crop, multi-layer cultivation diversifies harvest timing	Stabilizes income year-round and breaks the poverty cycle *1
Regional Development	Adoption of the method triggers local ecosystem regeneration	Improves working conditions and builds sustainable communities

*1 Helps reduce child labor tied to household income.

6-2. Combination of Locally Suitable Crops and Trees

■ Crops/Trees to Grow and Their Roles

Category	Role	Key Examples
Livelihood Improvement	Cash income from harvest and sales	Sweet potato, okra, cassava, vanilla, cacao, mango, papaya, avocado
Environmental Restoration	Soil improvement & CO_2 sequestration	Moringa, acacia, teak
Fuel	Fuel for biodiesel water trucks	Moringa, jatropha, sunflower, castor bean, rapeseed

Note: Crop and tree selection should respect local food culture and expert guidance.

6-3. Global Environmental Contribution of 10,000 Sites

40 ha × 10,000 sites could result in approximately 6 million tons of annual CO_2 sequestration.

Roughly comparable to the annual CO_2 emissions of Iceland.

*Model case; results vary by vegetation density and climate conditions.

Average estimates below use a 1,000-person settlement model (20–40 ha),

since carbon sequestration varies by crop/tree type.

Cultivated Area (per settlement)	Estimated Annual CO_2 Sequestration (per site)	Number of Projects	Estimated Total Annual CO_2 Sequestration
20 ha	200–400 tons	×10,000	2–4 million tons
30 ha	300–400 tons	×10,000	3–4 million tons
40 ha	400–800 tons	×10,000	4–8 million tons

The 6 million tons figure is an illustrative model estimate.

7. Application Sites

7-1. Target Settlements Requiring Deployment of This Integrated Model

- Africa: 54 countries

  Africa has a population of 1.4 billion; over 400 million people are estimated to lack access to safe water.

- African settlements

  300–1,500 people (most common)

  2,000–5,000 people (medium-sized)

  10,000+ people (treated as towns)

- Settlements with water issues

  Hundreds of thousands of settlements in Africa face water-related issues.

  Depending on the country, thousands to tens of thousands of settlements may be target sites.

- Estimated applicable areas

  A preliminary conceptual estimate is around 500,000 settlements.

7-2. Target Regions Requiring Deployment of This Integrated Model

- Applicable to Southeast Asian regions with water issues
  Bangladesh, Cambodia, Laos, Myanmar, parts of Indonesia, parts of the Philippines

- Applicable to other regions as well
  Desert and semi-arid regions across the Northern Hemisphere
  Potentially deployable in ways that consider water resources and ecosystems.

  Northern China
  Around Mongolia
  Central Asia (southern Kazakhstan, Uzbekistan, Turkmenistan)
  Middle East (Jordan, northern Saudi Arabia, western Iraq)

8. Other Considerations

8-1. Other Considerations: Do Not Harm

- **Must not become a source of conflict**

  Facilities built by this project must not become a flashpoint between the national government and anti-government armed groups.

  Local residents' understanding is needed to avoid any sense of unfairness between project areas and non-project areas.

8-2. Other Considerations: Implementation Sites

- **Implementation sites (solar panels, small-scale seawater desalination, agroforestry)**

  Land should be secured through state ownership or community-agreed land-use rights

- **Water allocation**

  How to decide allocation of drinking/domestic water, agricultural water, and water for growing forests.

- **Salt production**

  Concentrated brine discharged from small desalination/water purification units (when using seawater) can be used to produce salt.

  Though limited, it can help improve local livelihoods and may cover maintenance costs such as filters.

8-3. Other Considerations: IT-Related

- **Battery Selection: Avoid Conflict-Mineral Risks**

  Adopt cobalt-free LFP (lithium iron phosphate) batteries to reduce conflict-mineral and child-labor risks
  while maintaining safety in high-temperature environments.

  Note: Lithium-ion batteries may contain conflict minerals such as cobalt, and harsh child labor at mining sites is a concern.

  To reduce fossil-fuel dependence, avoid generators powered by fossil fuels.
  If desalination must operate after sunset, use electricity from biodiesel generators.
  Consider reducing the number of units; until large sodium-ion batteries become commercialized, purify as much water as possible during daylight hours and store it in tanks.

  Note: Biodiesel is a renewable fuel, but its lifecycle greenhouse-gas emissions depend on the feedstock, production process, transport, and land-use impacts.

8-4. Other Considerations: Operations

- **Build an IT-based remote monitoring system**

  Monitor power generation, detect desalination/purification unit anomalies, and alert for filter replacement.

  Because the units do not support remote monitoring, monitor equipment status with remote cameras.

  Note: Saving camera footage and temporarily storing logs will be designed during implementation.

- **Equipment maintenance**

  Regular maintenance is required for desalination/purification units and solar power systems.

- **Lightning rods**

  Always install lightning rods to prevent equipment failure caused by lightning strikes.

8-5. Other Considerations — Risk Avoidance: Desert Locust Countermeasures (Short / Medium / Long Term)

- Risks posed by desert locusts
  - Can devastate crops and forests in a short time
  - Gregarious swarms range from tens of millions to billions of individuals
  - Travel more than 100 km per day
  - The early stage of agroforestry is the most vulnerable

- Short term: Leverage FAO's early warning system
  - FAO Locust Watch: continuously monitor outbreak status, migration forecasts and alert levels.

- Medium term: Eliminate at the larval (hopper) stage (carried out at village level)
  - Larvae (hoppers) have no wings and only walk on the ground — this stage is the easiest to control
  - Methods: herd into trenches and bury, barrier traps, small-scale spraying (conventional spraying, not ULV)
  - Can be addressed at the village level → a potentially practical way to reduce the risk of major outbreaks

- Long term
  - Increasing forest cover
  - Natural predators (birds, insects, small animals)
  - Soil stabilizes and desertification is curbed → over the long term this reduces the very breeding grounds of desert locusts

9. What to Expect in the Future

9-1. Potential Natural Rainfall Feedback from Increased Vegetation

As a hypothesis for further study, increased vegetation and evapotranspiration could potentially contribute to changes in local water-cycle dynamics in some arid areas.

Under favorable regional climate and atmospheric conditions, changes in water-cycle dynamics may influence local rainfall patterns. In addition to planted forests, naturally regenerated vegetation could spread, potentially increasing CO_2 sequestration.

This could potentially contribute to a positive feedback loop that supports vegetation recovery and may help reduce desertification in some arid regions.

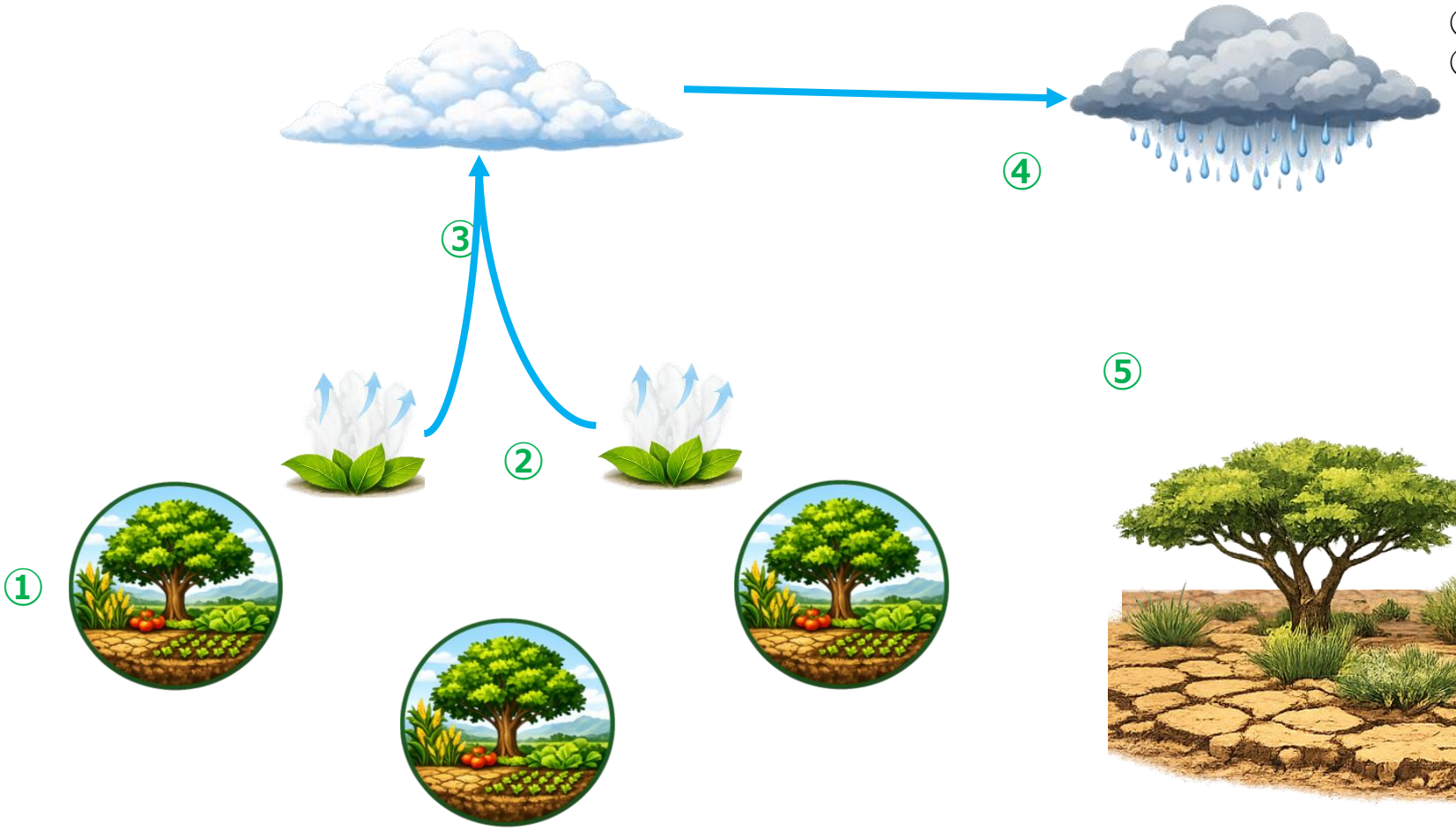
The specific implementation regions and forest areas should be determined by experts, but

for example, in an arid area about the size of Narita Airport (approx. 1,000 ha), restoring about 100–300 ha as forest could potentially influence local water-cycle dynamics

Under favorable regional conditions, this could potentially contribute to changes in local rainfall patterns and support water-cycle recovery in surrounding areas.

9-2. Natural Rainfall Mechanism — Conceptual Image

① Deploy multiple agroforestry sites
② Transpiration from forests
③ Increased evapotranspiration
→ atmospheric moisture and cloud formation
④ Natural rainfall
⑤ Vegetation establishes naturally



⑤



9-3. Crop Production + Forest Growth with Other Farming Methods

Where seasonal rainfall is sufficient

- **Half-moon farming**
  → Rainwater collects effectively in the basins

- **Zai farming**
  → Water and nutrients gather in the holes, sharply increasing germination rates

- **Stone lines**
  → Water flow slows, helping soil remain instead of being washed away

- **FMNR (Farmer Managed Natural Regeneration)**
  → A core technique for forest restoration, making forest regeneration easier

9-4. Reducing Hunger and Conflict

Through this model, water access, agriculture, and livelihoods may become more stable, potentially helping to reduce hunger.

Disputes over local resources and migration for livelihoods may decrease, potentially reducing the risk of "competition-driven conflict."

This not only satisfies the basic principle of international cooperation,

"Do No Harm (avoid causing harm)" but could also contribute to regional stability and peacebuilding.

10. Summary

10-1. Summary: Three Areas of Impact (1/3)

1. Water Security & Health:

Providing stable access to safe water and sanitation, reducing the burden of water collection, improving access to basic healthcare, and enabling educational opportunities.

These improvements can also contribute to reducing preventable maternal and newborn deaths.

2. Agroforestry & Ecosystem Regeneration:

Supporting economic growth, helping to reduce child labor associated with household poverty, and contributing to ecosystem regeneration.

3. Sustainable Community Building:

Creating infrastructure powered primarily by renewable energy, while reducing reliance on fossil fuels.

A Key Condition for Sustainable Implementation

Fair access to water and land, equitable benefit-sharing, and transparent governance involving local communities are essential for sustainable implementation.

10-1. Summary: Three Areas of Impact (2/3)

Standardization of the 1,000-Person Unit Model

■ **Unified Design:**

Standardized capacity, equipment, and renewable energy configuration for any region.

■ **Versatile Application:**

Fully compatible with both seawater and freshwater sources.

■ **High Scalability:**

Designed for broad implementation across diverse countries

10-1. Summary: Three Areas of Impact (3/3)

Scaling Potential & Vision

■ **Broad Reach:**

Applicable to hundreds of thousands of communities across Africa and Asia.

■ **Environmental Impact:**

Potentially achieving approximately 6 million tons of annual CO_2 sequestration through 10,000 sites.

■ **Social Impact:**

Improving health and quality of life for children and families through simultaneous water, sanitation, healthcare, and education improvements.

■ **Global Contribution:**

Proposing a Japan-led international model to address global challenges.

"Targeting Broad Deployment by 2040"

10-2. Relationship with the SDGs

I want to improve access to safe water, sanitation, and basic healthcare for children and families.



Improving livelihoods through crops

Escaping poverty through regional development



Securing food through crop harvests



Access to basic healthcare

Improved child and family health



Reduce the burden of water collection

Higher incomes reduce child labor and enable school attendance



Relieve women and girls of water-carrying duties

Education → better jobs → prevention of child marriage



Securing safe, hygienic water

Safe toilets with electric lighting



Use of renewable energy



Reduce Child Labor



CO_2 sequestration through tree planting and agroforestry



Forest growth

Crops through drip irrigation and mulching

10-3. Additional Potential Impacts

Issue	Method	Effect	SDGs
Nutrition / food insecurity	Diverse crops and improved livelihoods	Improved child health and development; improved food access	SDGs 2,3
Water collection burden	Access to safe and hygienic water	Women and children relieved from water-collection duties	SDGs 5,6
Maternal and newborn health	Safe water and sanitation; access to basic healthcare	Reduced risk of preventable maternal and newborn deaths	SDGs 3,6
Child labor	Safe water, agroforestry, improved livelihoods through crops	Reduced child labor undertaken to support household income	SDGs 8
Child marriage	Improved livelihoods and restored educational opportunities	Reduced risk of child marriage	SDGs 5
Nighttime safety for girls	Renewable-energy lighting, security cameras and patrols	Girls can safely use toilets at night	SDGs 5,6
Lack of nighttime learning opportunities	Renewable energy + lighting	Opportunities to study at night	SDGs 4,7
Energy shortage	Renewable energy	Improved access to lighting and water treatment	SDGs 7

10-4. Toward a Sustainable Cycle of Support

No single project can reach everyone in need. It is difficult to reach everyone through ODA and donations alone. Where this approach cannot reach, I hope NGO support can continue to provide assistance to those who need it most.

As more communities improve their livelihoods through this approach, I hope that NGO activities and other forms of external support can reach people and communities that still need assistance.

Looking further ahead, I believe an even more positive cycle could emerge if some of those whose lives have improved through support go on to support NGOs and others in need themselves.

As communities become more economically self-reliant, a portion of the benefits generated through improved livelihoods could be voluntarily reinvested in supporting other communities that still face similar challenges.

This could create a cycle in which communities that have benefited from support gradually become contributors to the development of the next communities.

10-5. Small-Start and Wider Application

This approach can start on a small scale.

This proposal uses a community of 1,000 people and 40 hectares of agroforestry land as a model. However, the scale can be adjusted flexibly according to the population and available agricultural land. This scalability is one of the strengths of the approach.

I learned about initiatives such as the Sahel Forest Project (SFP) and the Great Green Wall (GGW) after developing this proposal. I hope that this approach can be implemented not only in specific regions, but across many regions.

I would also be pleased if this approach could contribute, even partially, to securing water sources and supporting other projects such as the Great Green Wall and projects being implemented in Asia.

10-6. Community Development and Wider Benefits

- As livelihoods improve, communities can gradually expand solar power capacity to support further development.
- Biodiesel-powered vehicles can facilitate the transportation and distribution of agricultural products.
- As food, clothing, housing, and other basic needs become more stable, people can devote more time and resources to education, work, and other community activities.
- Where capacity allows, communities can share water and other resources with neighboring communities facing similar challenges.

10-7. Initial Funding, PoC and Path to Wider Deployment

- **Initial Funding**
  One of the key challenges for the initial deployment of this approach is the upfront investment required for infrastructure and equipment. In the early stages, **ODA and other public funding mechanisms** could be utilized to support initial implementation and demonstration.

- **From PoC to Wider Deployment**
  PoC is important for validating technical feasibility, local suitability, safety, and cost-effectiveness. However, this proposal combines existing concepts and technologies rather than relying on an entirely new technology. Therefore, rather than waiting for a large-scale PoC to be fully completed, a step-by-step approach of **small-scale implementation, testing, improvement, and expansion** could be considered.

- **Sustainable Financing for Wider Deployment**
  As the approach expands beyond initial projects, financing opportunities linked to climate-related policies and market mechanisms, including the **Joint Crediting Mechanism (JCM)** and, where applicable, **national ETS** frameworks such as Japan’s **GX-ETS**.

This proposal was created for **children of the future**.

I want to improve the health and well-being of children and families.

I want to reduce the number of children who must fetch water,
work to support their families, and give up school.

I want to help reduce atmospheric CO_2 and help reduce the risks children face from heat waves and droughts.

I want to create a future where, as adults, they can help build hope
for the next generation.

That is the spirit behind this proposal.

Thank you for your consideration.

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I am entrusting this to a promising young person.

22 Apr 2026 (First edition)

12 Sep 2026 (Updated)

Kenichi Miyauchi

mailto:miyauchi [dot] kenichi [at] outlook [dot] com

This proposal receives technical cooperation from Waqua Co., Ltd. for compact seawater desalination equipment (MYZ-E series).

*This document does not cover detailed specifications related to corporate confidential information.

AI Disclosure

AI-assisted tools, including ChatGPT, Microsoft Copilot, and Google Gemini, were used to support language editing, drafting assistance, and review during the preparation of this proposal.

The author reviewed and verified the outputs and takes full responsibility for the final content.

■ Revision History

Date	Changes
22 Apr 2026	First edition published
25 Apr 2026	Updated to Rev. 1.1; added information on microplastic removal
18 Jul 2026	Updated to Rev. 1.2; revised title and content. Updated the Japanese version and released the English version.
12 Sep 2026	Updated to Rev. 1.3; revised title and content. Updated the English and Japanese version. Updated to Rev. 1.3; updated for submission to the WHO call on Women's, Children's and Adolescents' Health beyond 2030